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A First-Order Matrix Approach to the Analysis of Electron Beam Emittance Growth Caused by Coherent Synchrotron Radiation

Ryoichi HAJIMA*

Advanced Photon Research Center, Japan Atomic Energy Research Institute, Tokai-mura, Ibaraki 319-1195, Japan

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A first-order transform matrix is proposed for calculating electron beam emittance dilution arising from coherent synchrotron radiation (CSR) in a next-generation light source based on an energy-recovery linac. The matrix approach enables us to scan numerous parameters for the design of achromatic cells of minimum emittance dilution. The emittance dilution can be minimized when the envelope of CSR wake dispersion matches the betatron function envelope at the achromatic cell exit. [DOI: 10.1143/JJAP.42.L974]

KEYWORDS: coherent synchrotron radiation, emittance, transfer matrix, energy-recovery linac, light source

An energy-recovery linac (ERL) is considered as a promising device for realizing a next-generation light source, which produces high-brightness and high-flux VUV and X-ray radiation with femtosecond pulse duration.[1)] Since the brightness of the radiation markedly depends on electron beam emittance, the analysis of beam emittance dilution in ERLs is one of the critical issues for the development of future ERL light sources. Quantum excitation by synchrotron radiation has been a major source of beam emittance dilution in storage ring light sources. In ERL light sources, quantum excitation is negligibly small, because an electron bunch circulates only once and goes to a beam dump after the circulation.[2)] The generation of an electron bunch as short as 100 fs, a unique property of ERLs, produces another source of emittance growth, that is, coherent synchrotron radiation (CSR).

An electron bunch emits CSR along a circular path such as a bending magnet. The emitted CSR power from each electron is a function of the longitudinal position of the electron in the bunch, which is expressed in terms of CSR wake potential.[3)] If we decompose the bunch into numerous slices along the longitudinal direction, each bunch slice has an independent error in bending angle due to the energy change resulting from the CSR emission. This CSR kick on a beam slice dilutes the projection emittance after the circular path. This CSR-induced emittance growth has been studied intensively in the design of electron bunch compressors for X-ray free electron lasers,[4)] which includes the development of analytical models[5)] and numerical simulation codes.[6,7)]

The electron beam transport path of an ERL light source, which consists of achromatic cells, is designed to keep the electron bunch length constant (isochronous) or to compress it gradually. In such a beam transport system, the CSR-induced emittance dilution can be estimated via an analytical method without time-consuming computations of particle tracking. We propose, in this Letter, a first-order matrix approach to the analysis of emittance dilution by CSR in ERL light sources. The matrix approach enables us to scan a wide range of design parameters of achromatic cells quickly and to find an optimum design for minimum emittance dilution.

We start with a first-order equation of electron motion in a uniform field of a dipole magnet:

$$x'' = -\frac{x}{\rho^2} + \frac{1}{\rho}[\delta_0 + \delta_{\mathrm{csr}} + \kappa(s - s_0)], \tag{1}$$

where x is the deviation of an electron from the ideal path in the bending plane, the derivative is taken with respect to a coordinate along the ideal path s, ρ is the bending radius of the ideal path, and δ_0 is the initial momentum deviation at $s = 0$ normalized by the reference energy. These terms are quite the same as those for the analysis of electron motion in a dispersive system.[8)] The last two terms on the right-hand side are introduced to calculate the CSR effect, in which δ_{csr} is the normalized momentum deviation caused by CSR in an upstream path ($0 < s < s_0$), $\kappa = W/E_0$ is the normalized CSR wake potential in the bending path and is defined by CSR wake potential W[eV/m] and reference energy E_0[eV], and $s = s_0$ is the entrance of the bending magnet.

We can attribute κ to each electron under the following assumptions: (1) all the dipole magnets have the same bending radius, (2) the electron bunch does not change its longitudinal profile in a single achromatic cell, and (3) the transient CSR effect at the entrance and exit of the magnet is not large, which are reasonable for ERL light sources. In this case, we have $\delta_{\mathrm{csr}}(s_0) = \kappa L_{\mathrm{b}}(s_0)$, where $L_{\mathrm{b}}(s_0)$ is the total bending path length for $0 < s < s_0$. Given a vector to specify the motion of an electron at $s = s_0$, $\boldsymbol{x}(s_0) = (x, x', \delta_0, \kappa L_{\mathrm{b}}, \kappa)^T$, as the initial condition, we can solve the electron trajectory through the bending magnet and specify the vector $\boldsymbol{x}(s_1)$ at the exit of the magnet using eq. (1). These two vectors can be also connected using a first-order transfer matrix:

$$\boldsymbol{x}(s_1) = R_{\mathrm{bend}}\boldsymbol{x}(s_0). \tag{2}$$

The transfer matrix for a sector magnet with bending angle θ and bending radius ρ is derived by Green's function method:[8)]

*E-mail address: hajima@popsvr.tokai.jaeri.go.jp

$$R_{\mathrm{bend}} = \begin{pmatrix} \cos\theta & \rho\sin\theta & \rho(1-\cos\theta) & \rho(1-\cos\theta) & \rho^2(\theta-\sin\theta) \\ -\frac{1}{\rho}\sin\theta & \cos\theta & \sin\theta & \sin\theta & \rho(1-\cos\theta) \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & \rho\theta \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \quad (3)$$

This is an extension of a 3x3 R-matrix used in beam transport analyses with momentum dispersion. Similar matrices are obtained for a quadrupole and a drift, and the electron motion along a beam transport system can be calculated from these R-matrices.

The off-axis motion of an electron caused by the initial momentum error is expressed in terms of the momentum dispersion function $\eta_x(s)$:

$$\left(\eta_x(s_1)\ \eta_x'(s_1)\ 1\ 0\ 0\right)^T = R_{0\to1}\left(\eta_x(s_0)\ \eta_x'(s_0)\ 1\ 0\ 0\right)^T. \quad (4)$$

We can define, in a similar manner, the CSR wake dispersion function $\zeta_x(s)$ to track the off-axis motion due to the CSR effect:

$$\left(\zeta_x(s_1)\ \zeta_x'(s_1)\ 0\ L_b(s_1)\ 1\right)^T = R_{0\to1}\left(\zeta_x(s_0)\ \zeta_x'(s_0)\ 0\ L_b(s_0)\ 1\right)^T, \quad (5)$$

and we find that an electron has a displacement $\boldsymbol{d} = (\kappa\zeta_x, \kappa\zeta_x')$ in the (x, x') phase space.

In the design of achromatic cells for a beam recirculation loop of ERLs, quadrupole parameters are chosen so that we can eliminate the dispersion due to the initial momentum spread at the exit of each cell, $\eta_x = \eta_x' = 0$. In this case, the dispersion by CSR wake has a nonzero value, $\zeta_x \neq 0$ and $\zeta_x' \neq 0$, at the cell exit. This residual dispersion induces the growth of projection emittance. Since the deviation of an electron due to the CSR wake is expressed as $(\kappa\zeta_x, \kappa\zeta_x')$ in the first-order approximation, each bunch slice aligns on the line $\zeta_x x' - \zeta_x' x = 0$ in the (x, x') phase space as shown in Fig. 1.

For the compensation of the CSR-induced emittance growth, two design strategies have been proposed.[4,9,10] One is applying two CSR kicks to cancel each other by inserting beam transport of "$R = -I$" between two cells or setting cell-to-cell betatron phase advance at an appropriate value. The other is matching the CSR kick to the orientation of the phase ellipse at the cell exit.

The R-matrix method suggested in this Letter can be adopted to the achromatic cell optimization for a minimum CSR effect. For an example of a triple-bend achromat (TBA), we can minimize the CSR effect by setting quadrupoles so that ζ_x and β_x have the same envelope after the cell. Figure 2 shows an optimized cell obtained by the R-matrix method, where the bending angle is $(3+6+3) = 12$ degrees, the bending radius is 25 m, and an isochronous condition is chosen. At the exit of the last bending magnet, Courant-Snyder parameters for the bending plane are $\alpha_x = -10.3$ and $\beta_x = 14.8$ m, which correspond to the phase ellipse orientation $\phi = 0.61$ rad. The CSR wake dispersion function at the last bend exit is found to be $(\zeta_x, \zeta_x') = (-0.388\,\mathrm{m}^2, -0.272\,\mathrm{m})$, that is $\phi_\zeta = 0.61$ rad. The direction of the CSR kick, thus, matches the orientation of the phase ellipse, and the emittance compensation is achieved.

The emittance compensation in the optimized TBA cell has been confirmed by the particle tracking code ELEGANT including CSR calculation.[11] We assume the initial electron bunch parameters as follows: central energy 3.07 GeV ($\gamma = 6000$), bunch charge $Q = 770$ pC, normalized rms emittance $\varepsilon_{n,0} = 0.1$ mm-mrad, longitudinal size $\sigma_s = 30\mu$m (100 fs), uncorrelated energy spread $\sigma_E/E_0 = 0.02\%$, and Gaussian distribution of electrons in the 6D phase space of electron motion. Projection emittance at the cell exit is calculated with scanning Courant-Snyder parameters at the cell entrance: $-5.0 < \alpha_x < 3.0$, $\gamma_x = 0.29\,\mathrm{m}^{-1}$. In these calculations, one million particles are tracked with a 1cm step. Figure 3 shows the calculated projection emittance as a function of phase ellipse orientation angle at the last bend exit. It has been confirmed that the projection emittance becomes minimum, $\varepsilon_n = 0.178$ mm-mrad, when the phase ellipse angle coincides with the CSR kick direction $\phi_\zeta = 0.61$ rad.

The amount of the emittance growth for the optimized

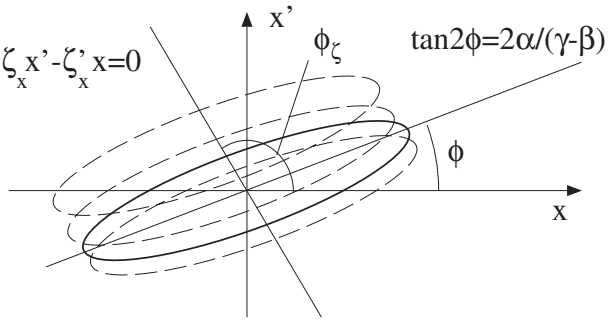


Fig. 1. Two-dimensional phase ellipse in the (x, x') plane, and displacement of beam slices due to the CSR kick. (α, β, γ) are Courant-Snyder parameters.

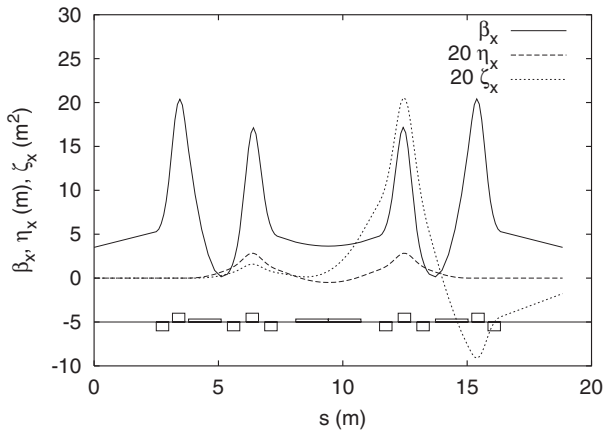


Fig. 2. Betatron function β_x, momentum dispersion η_x and CSR wake dispersion ζ_x of an optimized triple-bend achromat. The envelope of the CSR wake dispersion matches the betatron envelope after the last bending magnet.

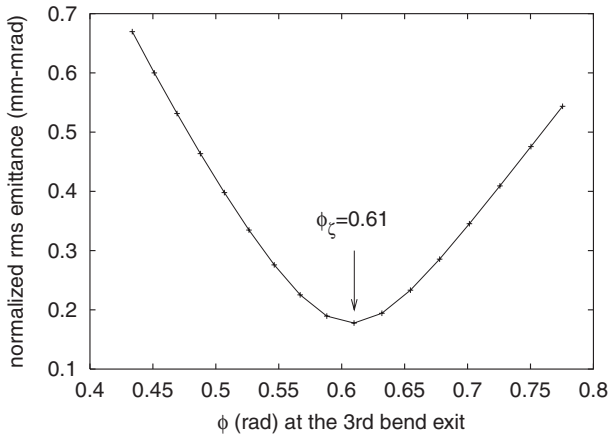


Fig. 3. CSR-induced projection emittance growth for various phase ellipse parameters calculated by particle tracking. The horizontal axis is the orientation of the phase ellipse at the last bend exit. The initial normalized emittance is $\varepsilon_{n,0} = 0.1$ mm-mrad. The direction of the CSR kick at the last bend exit is $\phi_\zeta = 0.61$ rad.

achromatic cell can be estimated by simple calculation instead of by particle tracking calculation. For a bunch with Gaussian distribution, the rms energy spread due to the CSR effect is given by[3)]

$$\Delta E_{\mathrm{rms}} \simeq 0.22 \frac{eQL_b}{4\pi\epsilon_0 \rho^{2/3} \sigma_s^{4/3}}. \tag{6}$$

Using the above calculation parameters, we have $\Delta E_{\mathrm{rms}} \simeq 990\,\mathrm{keV}$, which is equivalent to the spread of the CSR wake potential $\Delta W_{\mathrm{rms}} = 190\,\mathrm{keV/m}$, $\Delta\kappa_{\mathrm{rms}} = \Delta W_{\mathrm{rms}}/E_0 = 6.2 \times 10^{-5}/\mathrm{m}$. This CSR wake potential results in the nonuniform displacement of bunch slices in the (x, x') phase space. When the achromatic cell is optimized for the envelope matching between the CSR-induced dispersion function and the betatron function, all the bunch slices align along the orientation of the phase ellipse and have a distribution of the displacement $(\Delta\kappa_{\mathrm{rms}}\zeta_x, \Delta\kappa_{\mathrm{rms}}\zeta_x') = (24\,\mu\mathrm{m}, 17\,\mu\mathrm{rad})$. The rms beam size at the same position without the CSR effect is calculated from the betatron function and the emittance to be $\sigma_x = 15.7\mu\mathrm{m}$. The growth of the rms projection emittance due to the CSR effect is estimated from $\varepsilon_n/\varepsilon_{n,0} = \sqrt{[(\Delta\kappa_{\mathrm{rms}}\zeta_x)^2 + \sigma_x^2]/\sigma_x^2} = 1.8$, which is consistent with the result from particle tracking shown in Fig. 3.

In conclusion, we have derived a first-order transform matrix for calculating electron beam dynamics including emittance dilution arising from coherent synchrotron radiation (CSR). Since the matrix approach enables us to scan numerous design parameters quickly including the CSR effect, it can be adopted to the design optimization of achromatic cells for ERL light sources. It has been shown that the dilution of the projection emittance due to the CSR can be minimized by establishing the envelope matching between the CSR-induced dispersion function and the betatron function, and the amount of the emittance dilution is also estimated analytically. The matrix approach can be used for both unshielded and shielded CSR with an arbitrary bunch shape, when the CSR wake potential is given. Although we limit our analyses to achromatic cells with a constant bending radius in this Letter, the matrix approach can be extended to the case of nonuniform bending radii with a minor modification. Particle tracking calculation is necessary to refine the cell design including higher-order beam dynamics and CSR effects in a transverse direction and/or a transient regime. The matrix approach is still helpful even in this case because we can prescreen the design parameters by the matrix method before applying time-consuming particle tracking.

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